

## RESEARCH ARTICLE

JASIST WILEY

# Differential impacts of social influence on initial and sustained participation in open source software projects

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## Funding information

Guangdong Province Focus Research Project, Grant/Award Number: 2019KZDZX2014; Guangdong Province Soft Science Fund, Grant/Award Number: 2020A1010020002; National Natural Science Foundation of China, Grant/Award Numbers: 71971106, 72001099; Shenzhen Humanities & Social Sciences Key Research Bases; Starting Scientific Research Funding from Beijing Foreign Studies University, Grant/Award Number: 2019QD027

## Abstract

Social networking tools and visible information about developer activities on open source software (OSS) development platforms can leverage developers' social influence to attract more participation from their peers. However, the differential impacts of such social influence on developers' initial and sustained participation behaviors were largely overlooked in previous research. We empirically studied the impacts of two social influence mechanisms—word-of-mouth (WOM) and observational learning (OL)—on these two types of participation, using data collected from a large OSS development platform called Open Hub. We found that action (OL) speaks louder than words (WOM) with regard to sustained participation. Moreover, project age positively moderates the impacts of social influence on both types of participation. For projects with a higher average workload, the impacts of OL are reduced on initial participation but are increased on sustained participation. Our study provides a better understanding of how social influence affects OSS developers' participation behaviors. It also offers important practical implications for designing software development platforms that can leverage social influence to attract more initial and sustained participation.

## 1 | INTRODUCTION

The success of an open source software (OSS) project largely relies on voluntary participation from its developers (Fang & Neufeld, 2009; Roberts et al., 2006). Major OSS development platforms, such as GitHub and OpenHub, have implemented various social functions and strategies (e.g., microblogging and information display strategies) to leverage developers' social influence to attract participation from their peers, including two major social influence mechanisms—word-of-mouth

(WOM) and observational learning (OL)—that operate quite differently in terms of changing others' opinions or behaviors. Thus, an important research challenge is to discover whether action (OL) speaks louder than words (WOM) in terms of influencing OSS developers' participation or vice versa.

Moreover, it was found that OSS developers' initial participation (i.e., contributing to a project for the first time) and sustained participation (i.e., continuous contribution) are often driven by different forces (Fang & Neufeld, 2009). To more accurately leverage WOM and

OL, it is important to better understand their differential impacts on OSS developers' initial and sustained participation.

We suggest that WOM and OL attract OSS project participation differently. WOM refers to the process of influencing the behaviors of a person's socially connected peers via communications of opinions or recommendations (Arndt, 1967; Chen et al., 2011). In the setting of our empirical study—a large OSS development platform called OpenHub, developers use a microblogging service to share information, such as their development experiences, opinions or recommendations, with their follower developers. We suggest that this microblogging service serves as the channel of WOM for a developer who influences the participation of her followers. As Table 1 shows, message 8478 and 8491 directly call the followers to contribute. Message 1361 shows positive opinions about a project and may attract followers to become interested and participate. Some social interactions, such as message 43477 and 64799, express microbloggers' appreciations of their peers, and thus can motivate them to increase their contributions. Thereby this microblogging service allows follower developers to easily access microbloggers' subjective opinions or recommendations about OSS projects.

On the other hand, OL in this study is defined as developers observing and being influenced by others'

project participation behaviors. Such objective information plays an important role in an OSS developer's project participation decisions. Figure 1 shows an example of OL information on OpenHub: the 30-day and 12-month development activities of an OSS project “Chamilo” in 2017. A developer may notice the relatively low total commits in 2017 and the 16% drop from last year and decide not to contribute any more if at all.

The goal is to study the differential impacts of the WOM and OL social influence mechanisms on developers' participation decisions, more specifically, to answer the two following research questions:

**RQ1.** What are the impacts of WOM and OL on OSS developers' initial project participation?

**RQ2.** What are the impacts of WOM and OL on OSS developers' sustained project participation?

This study has three main contributions. First, this study aims to explore the differential impacts of WOM and OL mechanisms on OSS developers' participation. In particular, why is one social influence mechanism specific to initial rather than sustained project participation? Second, our study provides possible explanations of the differential impacts from the information accessibility and diagnosticity perspective. Last, the empirical findings can help OSS stakeholders devise practical strategies and design better social functions in development platforms to attract more participation, thereby improve project success.

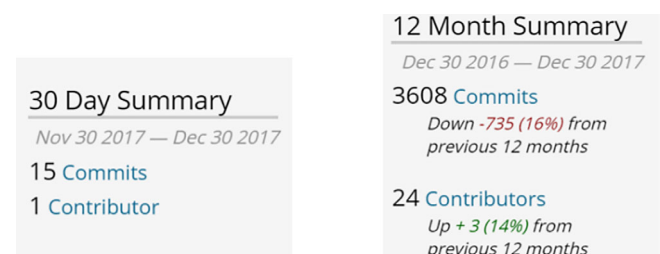
**TABLE 1** Samples of OpenHub microblogging messages

| Message_id | Created_time    | Content                                                                                                                 |
|------------|-----------------|-------------------------------------------------------------------------------------------------------------------------|
| 1361       | 2008/6/7 11:01  | Went from ~100 failures in the gigantic Expression Tree test to 3. Linq will sure look good in #mono 2.0.               |
| 8478       | 2008/10/2 16:50 | Looking for help! Please contact us at: flowdesigner-devel at lists dot sourceforge dot net. Any contribution welcomed! |
| 8491       | 2008/10/2 21:17 | We have worked in a new website for @ArgoUML! Enjoy it! Any improvements suggestion is welcome!                         |
| 43477      | 2009/8/1 20:27  | Finally made time to dig into #phpQuery. Incredibly worth the time. Great work @TobiaszCudnik                           |
| 64799      | 2010/1/23 22:51 | Starting development of #chamilo 1.8.7 with full support of UTF-8 thanks to @ivantcholakov                              |

## 2 | RESEARCH BACKGROUND

### 2.1 | Motivations for initial and sustained participation in OSS projects

Previous OSS research studied various motivation factors for developers' participation. While most of these studies



**FIGURE 1** OL information: developers' participation statistics of project “Chamilo” [Color figure can be viewed at wileyonlinelibrary.com]

do not explicitly distinguish these factors for initial or sustained participation, some of them discussed motivational factors for a developer to participate in a specific OSS project that we can categorize as initial participation. In subsequent subsections, we review OSS research on initial and sustained participation motivations that is related to social influence mechanisms and factors.

A common framework used in many previous OSS participation studies is to categorize participation motivations into two types: intrinsic motivations and extrinsic motivations (Li et al., 2012; von Krogh et al., 2012). Intrinsic motivations are often related to developers' personal needs, such as altruism and enjoyment, while extrinsic motivations are often about utility-related external rewards, such as learning and career advancement opportunities. These two types of motivations were found to influence both initial and sustained participation. For instance, Davidson et al. (2014) found that intrinsic motivations like altruism drove elder OSS developers' initial participation. For sustained participation, Hertel et al. (2003) found that intrinsic motivations like enjoyment and fun drive developers' sustained project participation. Stewart and Gosain (2006) discovered that OSS developers' adherence to the open source ideology affected the degree to which they were devoted to the underlying project. Moreover, Shah (2006) found that developers' initial participation decisions are often influenced by extrinsic motivations like software use value and learning opportunities, while their long-term contribution is increased over time for intrinsic motivations such as fun and enjoyment.

### 2.1.1 | Initial participation

A stream of studies on initial participation motivations related to the observation learning mechanism we introduced before is about developers' preferences for certain project characteristics. Such preferences often lead to preferential attachment phenomenon where developers favor projects with certain accumulative advantages (Capocci et al., 2006; Johnson et al., 2014; Kuk, 2006). For instance, Kuk (2006) suggested that developers were often attracted to participate in the most successful OSS projects (Maday et al., 2004) and often chose to work with peers with higher status indicators such as reputation and membership levels (Johnson et al., 2014). Qiu et al. (2019) found that OSS projects with more stars and commits are more likely to attract new participants. This finding is consistent with the OL mechanism in which developers may observe others' initial participation choices for a project. Such observation may arouse the observer's interests and willingness to work on this

project (Moqri et al., 2018). Other important project characteristics that may attract initial participation include project programming language (Casalnuovo et al., 2015) and prior social collaboration with existing project members (Casalnuovo et al., 2015).

### 2.1.2 | Sustained participation

One of the most important social factors that significantly affect developers' sustained participation is repeated (social) interactions among developers in which social influence is embedded. Wiertz and de Ruyter (2016) found that individuals' tendency to interact with people online positively affects their knowledge contribution behaviors. In OSS communities, this means developers who interact more with existing project members are more likely to sustain their participation. von Hippel and von Krogh (2003) suggested that the momentum for developers' sustained participation may be largely due to their social interactions with other team members. Shah (2006) found that long-term project participants enjoyed interacting with each other, and interactive discussion directly drives sustained participation (Daniel & Stewart, 2016). Moreover, Bagozzi and Dholakia (2006) found that engaged OSS participants often view their sustained contributions as "enjoyable joint activities to be done" with their peers. Kuk (2006) also suggested that encouraging developers' preferences in interacting with the most resourceful developers can help sustain their participation.

Other social factors that may affect an individual's sustained participation include her community experience (Fang & Neufeld, 2009), the attention she received (Moqri et al., 2018), her peer network size (Zhou & Mockus, 2012), value congruence (Maruping et al., 2019), and project characteristics, such as project governance (Tullio & Staples, 2013), license choices (Colazo & Fang, 2009), leadership style (Oh et al., 2016), software structure (Daniel & Stewart, 2016), project accreditation and code acceptance (Ho & Rai, 2017), relatedness, self-efficacy, and self-identity verification (Kuem et al., 2020).

### 2.1.3 | Initial versus sustained participation

Previous research suggested that motivations for initial and sustained participation can vary (Shah, 2006; von Krogh et al., 2012; Fang & Neufeld, 2009). von Krogh et al. (2012) stated that "... motivations change with the context of the social practice, a first entry into a community by making a contribution to OSS is motivated

differently from a sustained contribution to the same practice.” It was found that certain motivations that affect initial participation cannot affect OSS developers’ sustained participation (Fang & Neufeld, 2009; Shah, 2006). For instance, Fang and Neufeld (2009) found that project (information) access is a precursor to developers’ initial participation but is not sufficient to sustain participation, if not positively reinforced by other factors such as community recognition. Moreover, one of the developers’ initial participation motivations is use value (Fang & Neufeld, 2009; Shah, 2006). However, once a developer’s use need (as the initial motivation) is met, without other support factors like repeated social interactions, participation may decrease over time (Fang & Neufeld, 2009; Shah, 2006). Shah (2006) pointed out that some of the participation motivations may emerge only after developers’ initial participation. This finding highlights the importance of examining how initial participation and sustained participation are motivated differently, especially from the social influence perspective.

In summary, the mechanisms behind OSS developers’ motivations for initial and sustained participation may be quite different. Our study hypothesizes that different social influence mechanisms may have differential impacts on developers’ initial and sustained project participation.

## 2.2 | Impacts of social influence on OSS project participation

Social influence refers to the phenomenon that an individual’s behaviors can be influenced by her peers’ behaviors (Aral, 2011). It has been well studied in economics and marketing (Arndt, 1967; Bandura & Walters, 1977). Deutsch and Gerard (1955) categorized social influence into two types based on the two psychological needs for individuals to imitate others’ behaviors: (a) the need to be correct—informational social influence, and (b) the need to be liked—normative social influence. Informational social influence enables an individual to imitate her peers in order to reflect correct behaviors in ambiguous situations under uncertainty. Normative social influence drives an individual to conform to other persons to gain social approval. This conformity originates from one’s social needs for companionship and association. In an OSS development community, a developer may imitate others’ participation behaviors to fit in. However, this imitation may just be a public exhibition of compliance instead of private acceptance. In contrast, informational social influence often results in not only public compliance but also private acceptance (i.e., genuinely believing

that others are situationally correct). This categorization suggests that different types of social influence mechanisms are associated with different individual motivations, and thus, may induce different behaviors.

### 2.2.1 | Word-of-Mouth versus observational learning

While classical social influence studies focused on human psychological processes, marketing research mainly focused on studying the mechanisms that can leverage social influence to affect consumers’ purchase choices. Two mechanisms, word of mouth and observational learning were found to be particularly prominent in influencing individuals’ behaviors via social media and networking platforms such as Twitter and Facebook. In the OSS domain, Blincoe et al. (2016) suggested that it is important to understand how (social) influence is exhibited on social software development (i.e., social coding) environments. These environments have increasingly relied on social functions like microblogging that can facilitate WOM and OL to affect OSS developers. Our study then focuses on the impacts of WOM and OL on OSS developers’ participation behaviors.

**Word-of-Mouth** can be defined as the mechanism that individuals’ behaviors are influenced by others’ opinions and experiences through various communication channels (Arndt, 1967). Nowadays, these channels are often social media or online review web sites. Existing WOM studies mainly focused on its impacts on consumer behaviors and product sales (Cheung & Thadani, 2012). These studies found that WOM valence (positive or negative) can change consumers’ perception of a product (Mizerski, 1982), while WOM volume helps to raise consumer awareness of the underlying product and increase the number of informed consumers. Moreover, the emergence of online product reviews has attracted many scholars to study the impacts of WOM on consumers in the digital age. For instance, Clemens et al. (2006) conducted a survey of online reviews from the craft beer industry and found that products with high valence reviews are likely to be purchased again.

However, the impacts of WOM have been understudied in OSS research. Nakakoi et al. (2002) suggested that a new OSS developer often starts as a “lurker” by watching others’ OSS related interactions and then becomes more involved in OSS project participation. Bagozzi and Dholakia (2006) and Barbagallo et al. (2008) briefly mentioned that WOM can be useful in advertising OSS projects and building awareness among developers. Santos et al. (2013) suggested that WOM may have great potential in influencing developers’ participation behaviors.



More recently, Krishnamurthy et al. (2016) observed that OSS projects that receive more coverage in technology news websites are often associated with more peripheral developer participation. In summary, the impacts of WOM on OSS developers' participation behaviors were not systematically studied before.

Our empirical analysis collected data from a large OSS development community with a microblogging service. We hypothesize that developers may use microblogging to post their experiences, opinions, or activities to raise project awareness and attract more participation via WOM effects. This is consistent with the findings in Jansen et al. (2009) and Hennig-Thusrau et al. (2015), which suggest that microblogging offers a novel channel for WOM. Dabbish et al. (2012) and Seebach et al. (2011) also suggested that microblogging can enhance transparency and collaboration for software developers.

**Observational learning** refers to the social mechanism that individuals observe and are influenced by others' actions without knowing their actual motives (Bandura & Walters, 1977; Barbagallo et al., 2008). Previous research mainly studied the impacts of OL on consumers' product adoption decisions. When there is limited product information, the signals derived from publicly observed consumer adoptions often outweigh an individual's own private information in her adoption decision. As more and more consumers follow their peers' product adoptions, a behavior "herding" occurs. Moreover, the impacts of OL are often amplified in online environments, as individuals' web profiles and activities are becoming increasingly transparent (Cheung et al., 2015). For instance, Burke et al. (2009) found that social networking site users who observe their friends' content contributions are motivated to share more of their own. However, the impacts of OL were understudied in OSS research, while developers' participation and contribution activities become increasingly visible on various development platforms.

Hahn et al. (2008) found that OSS developers tend to participate in projects with team members they have previously worked with in other projects. OL mechanism may act an important role in this phenomenon. Researchers found that developers observe others' participation decisions and tend to join most popular projects (Dabbish et al., 2012; Tsay et al., 2014a; Tsay et al., 2014b). Singer et al. (2013) studied two developer profile aggregator sites and found that developers assess each other to find if others are worth following or worth collaborating with. It was also found that development activity information and developer profiles may attract more participation.

Sheoran et al. (2014) studied GitHub watchers (i.e., developers who watch repositories to receive notifications about project activity) and their contributions to the projects they watched. They found that projects with more watchers often can recruit more future contributors. Furthermore, they indicated that watchers are more likely to yield to sustained contributions than other contributors. Steinmacher et al. (2015) investigated social barriers that affect newcomers of OSS communities and found that it is necessary to "place the proper signs and maps to help newcomers orient themselves, and alert them about the obstacles that they might face," suggesting that OL can be helpful in reducing obstacles for developers' participation decisions.

Comparing WOM and OL mechanisms, the information conveyed through WOM are more of subjective (personal) opinions or evaluations. Publicly observed OL information often consists of objective statistics of developer activities. These differences in the objectivity of OL and WOM information may have differential impacts on developers' project participation. Moreover, OSS project participation is a process in which developers become more engaged, and thus, more familiar with the project and its members. We then also hypothesize that the impacts of WOM and OL may change during this process as individuals can better evaluate the project and its members' words or behaviors over time.

### 3 | RESEARCH DESIGN

The research design shown in Figure 2 consists of two major steps. The first step involves two components: (a) WOM and OL information extraction and (b) model specification. We extracted WOM and OL information from the microblogging messages and project profiles on Open Hub project homepages. The second step, econometric analysis uses the fixed effects model to analyze the impacts of WOM and OL on both initial and sustained project participation. We further elaborate each step in the following sections.

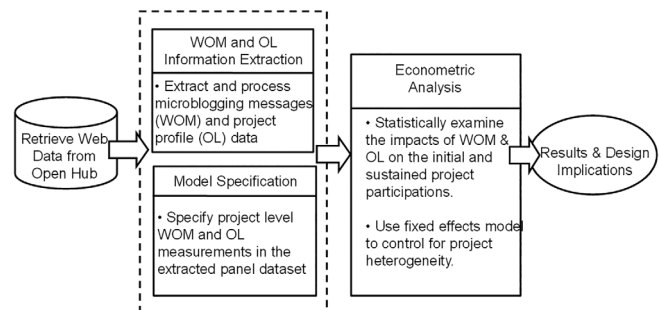


FIGURE 2 Research design of the empirical study

### 3.1 | Data

The data used in our study was collected from a large online open source software (OSS) development platform called Open Hub. It retrieves and analyzes publicly available information about OSS projects and developers from public software version control systems. Open Hub contains projects, ranging from well-known projects like MySQL to lesser-known ones like CakePHP. As of 2017, it contains more than 669,000 projects and 3.85 million developers, ranging from well-known projects such like MySQL to lesser-known ones like CakePHP. Moreover, Open Hub keeps track of the changes in the source code of OSS projects in the version control systems and calculated monthly software metrics such as the total number of commits, total number of participating developers, etc. These longitudinal development data enable us to construct a panel data sample for our econometric analysis.

We developed a set of Java programs to automatically retrieve data via the API of Open Hub. All data items retrieved were in XML format and parsed into a database. Such information includes OSS developers' project participation records, locations, nationalities, programming language preferences, and development activities.

Open Hub provided a microblogging service on its platform since 2008. This service allowed developers to publish their opinions, recommendations, and project participation activities to their followers, thereby may influence others' participation through such WOM social influence.

Moreover, the profile page of each Open Hub project displays the development information as Figure 1 shows, such as total number of commits, number of monthly new commits, etc. Therefore, a developer who views such OL information—her peers' participation activities—may eventually be influenced to participate or continue to contribute to this project.

Our empirical study collected development information about 1,716 Open Hub projects as well as their 7,881 participants from 2008 to 2009. Each of these projects had developers who adopt the microblogging service since 2008. Among the participated developers, 1,144 of them have posted 3,301 Microblogging messages through Open hub. We also extracted 22,308 records of these projects' monthly development activities from 2008 to 2009 to construct a panel data set for our econometric analysis.

### 3.2 | Model specification

Our empirical study aims to analyze how WOM and OL may affect OSS developers' initial and sustained project participation differently. We leverage the panel structure

of the extracted Open Hub data to control for the unobserved heterogeneity in project characteristics. We describe the detailed specifications of our fixed effects regression models as follows.

#### 3.2.1 | Dependent variables

The extracted Open Hub data contains monthly project development information as well as developers' microblogging and participation records. Therefore, in month  $t$ , we use the number of developers who contribute for the first time to a specific project  $i$  (**Initial\_zParticipation<sub>it</sub>**) as the dependent variable for measuring project  $i$ 's initial participation at time  $t$ .

On the other hand, we use the number of existing active developers (excluding first-time participating developers) who made at least one contribution to project  $i$  in month  $t$  (**Sustained\_Participation<sub>it</sub>**) as the dependent variable for measuring project  $i$ 's sustained participation.

#### 3.2.2 | Independent variables

**Word-of-Mouth (IP\_Microblogging\_Tags<sub>it-1</sub> and SP\_Microblogging\_Tags<sub>it-1</sub>):** The microblogging service of the OpenHub platform has provided a possible channel for OSS developers to influence their followers via textual information. Such information often contains microbloggers' opinions and recommendations about projects, as shown in the example in Table 1. As defined and discussed in the introduction, these messages may attract initial or sustained OSS project contributions from followers through WOM effects.

To analyze the impacts of WOM on initial participation, we use the number of microblogging messages that contain a project  $i$ 's tag in month  $t-1$  and read by at least one follower who has not participated in  $i$  before (**IP\_Microblogging\_Tags<sub>it-1</sub>**) as the independent variable. This measure ensured that this WOM channel was not preaching to the converted (i.e., developers who were  $i$ 's participants already and thus subscribed to microblogging messages containing  $i$ 's tag).

Similarly, for the analysis of sustained participation, we use the number of microblogging messages that contain a project  $i$ 's tag in month  $t-1$  and were read by at least one follower who has participated in  $i$  before (**SP\_Microblogging\_Tags<sub>it-1</sub>**) as the independent variable. This measure ensures that this WOM channel was preaching to project  $i$ 's existing members for sustained participation.

Moreover, prior social influence studies often model both the volume and the valence of WOM messages.

We have analyzed the valence of Open Hub microblogging messages using sentiment analysis and found that there is a relatively small percentage of negative messages about OSS projects. Therefore, our model only includes the volume of microblogging messages.

**Observational Learning (Monthly\_New\_Commits<sub>it-1</sub>):** To model the OL mechanism, we follow prior OL studies (Chen et al., 2011; Yi & Davis, 2003) that used the information about people's aggregated actions that are visible to the individuals who may be influenced. In the Open Hub platform, each project has a profile homepage that displays information about existing members' development activities from the last month. The information item that we adopt to model OL is the number of commits of a project in month  $t - 1$ . One commit is a one-time contribution of a developer to a project. Thus, we use Monthly\_New\_Commits<sub>it-1</sub> as the independent variable for OL.

### 3.2.3 | Control variables

**(OSS\_Experience<sub>it-1</sub>):** This variable is defined as the average amount of OSS development experiences (measured in months) for the members of project  $i$  in month  $t - 1$ . Roberts et al. (2006) used this variable as a proxy for developers' knowledge and skill that is often difficult to measure. The assumption is that more experienced developers may be able to attract more initial and sustained participation from developers who admire their skills and would like to learn from them.

**(Developers\_Reputation<sub>it-1</sub>):** Previous research found that developers with good reputation tend to attract more initial project participation (Hu et al., 2012). Newly joined developers may want to learn from senior project members about programming skills. In Open Hub, developers send kudos to each other to express their appreciations, and these kudos build up the receiver's community reputation. Projects with more reputable members tend to attract more participation (Hu et al., 2012). Therefore, we use the average number of kudos received by the developers of the OSS project  $i$  until month  $t - 1$  to control the average reputation status of  $i$ 's members.

**(Avg\_Commits<sub>it-1</sub>):** This variable is calculated as the average number of commits that project  $i$ 's members have made until month  $t - 1$ . This variable represents the average level of efforts and project knowledge (familiarity) of  $i$ 's members. Previous social influence studies show that an individual's knowledge level about a product can moderate the effects of social influence on product diffusion. The main assumption is that as an individual knows more about the product, she is less likely to be influenced by her peers. We then conjecture

that this variable may moderate the impacts of WOM and OL on sustained contribution.

**(Project\_Age<sub>it-1</sub>):** It was also found that project age may significantly affect developers' willingness for initial participation (Grewal et al., 2006; Hu et al., 2012). Hahn et al. (2008) suggested that different types of developers may prefer joining projects at different stages. Subramaniam et al. (2009) suggested that project age may be a proxy for other factors, such as group experience, and may affect project participation. Moreover, WOM and OL information may change as projects age. Therefore, we used project  $i$ 's age (measured in months) until month  $t - 1$  as both a control variable and a moderating variable for social influence.

**(Project\_Size<sub>it-1</sub>):** Following (Fang & Neufeld, 2009), we suggested that project size may influence the degree of OSS developers' engagement and commitment to the project, thereby affecting OSS developers' participation. We use the cumulative number of developers in project  $i$  until month  $t - 1$  as a control variable.

**(License<sub>i</sub>):** We control the license type as whether the project uses restrictive licenses such as GNU and GPL. Stewart et al. (2006) suggest that license restrictiveness affects OSS developers' project participation. They found that developers are attracted to projects that employ nonrestrictive licenses. Our analysis uses a dummy variable License<sub>i</sub>. This variable equals 1, indicating that project  $i$  uses a restrictive license.

**(Initial\_Participation<sub>it-1</sub>)** and **(Sustained\_Participation<sub>it-1</sub>)** are two lagged variables of dependent variables. These variables help address the temporal dependence issues in the data. Table 2 summarizes the detailed specifications of the previously described variables.

Our regression models control the project-specific fixed effects on unobserved idiosyncratic characteristics, such as programming languages and managers. Controlling these project-specific effects helps us to better identify the causal relationship between social influence and both types of project participation.

This approach is widely used in empirical research in economics and social science, where longitudinal observations exist for same subjects over time. The Durbin-Wu-Hausman test was conducted (Wooldridge, 2012) to determine whether the fixed effects model should be used, and the results of Hausman test indicated that the fixed effects model should be used. VIFs for the variables are lower than 10, indicating that independent variables do not suffer from multicollinearity issues. The detailed specification of our fixed effects models (without the interaction effects) are as follows:

We extended these two models by adding interaction terms between the independent variables (WOM and OL) and two possible moderators: Project\_Age<sub>it-1</sub> and

TABLE 2 Specifications of variables in the regression model

| Variables                                   | Descriptions                                                                                                                                                                      |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dependent variables</b>                  |                                                                                                                                                                                   |
| Initial_Participation <sub>it</sub>         | The number of newly joined developers of project <i>i</i> in month <i>t</i>                                                                                                       |
| Sustained_Participation <sub>it</sub>       | The number of existing project members who are actively contributing to project <i>i</i> in month <i>t</i>                                                                        |
| <b>Independent variables</b>                |                                                                                                                                                                                   |
| IP_Microblogging_Tags <sub>it-1</sub> (WOM) | The number of the microblogging messages that contain project <i>i</i> 's tag in month <i>t</i> - 1 and read by at least one follower who has not participated in <i>i</i> before |
| SP_Microblogging_Tags <sub>it-1</sub> (WOM) | the number of the microblogging messages that contain project <i>i</i> 's tag in month <i>t</i> - 1 and read by at least one follower who has participated in <i>i</i> before     |
| Monthly_New_Commits <sub>it-1</sub> (OL)    | The number of new commits made to project <i>i</i> in month <i>t</i> - 1. This information is displayed in <i>i</i> 's Open Hub profile page.                                     |
| <b>Control variables</b>                    |                                                                                                                                                                                   |
| OSS_Experience <sub>it-1</sub>              | The average level of OSS development experiences (measured in months) project <i>i</i> 's members have in month <i>t</i> - 1                                                      |
| Developers_Reputation <sub>it-1</sub>       | The average number of kudos received by project <i>i</i> 's members until month <i>t</i> - 1                                                                                      |
| Avg_Commits <sub>it-1</sub>                 | The average number of commits project <i>i</i> 's members have made until month <i>t</i> - 1                                                                                      |
| Project_Age <sub>it-1</sub>                 | The number of months project <i>i</i> existed until month <i>t</i> - 1                                                                                                            |
| Project_Size <sub>it-1</sub>                | The cumulative number of developers that project <i>i</i> has until month <i>t</i> - 1                                                                                            |
| License <sub>i</sub>                        | A dummy variable is used to measure the restrictiveness of <i>i</i> 's license. 1 indicates that project <i>i</i> uses a restrictive license.                                     |
| Initial_Participation <sub>it-1</sub>       | One-month lagged variable of the dependent variable for initial participation for controlling temporal dependence                                                                 |
| Sustained_Participation <sub>it-1</sub>     | One-month lagged variable of the dependent variable for sustained participation for controlling temporal dependence                                                               |

$$\begin{aligned} \text{Initial\_Participation}_{it} = & \alpha_1 \text{IP\_Microblogging\_Tags}_{it-1} + \alpha_2 \text{Monthly\_New\_Commits}_{it-1} + \alpha_3 \text{OSS\_Experience}_{it-1} \\ & + \alpha_4 \text{Developers\_Reputation}_{it-1} + \alpha_5 \text{Avg\_Commits}_{it-1} + \alpha_6 \text{Project\_Age}_{it-1} \\ & + \alpha_7 \text{Project\_Size}_{it-1} + \alpha_8 \text{License}_i + \alpha_9 \text{Initial\_Participation}_{it-1} + \rho_i + \varepsilon_{it} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{Sustained\_Participation}_{it} = & \beta_1 \text{SP\_Microblogging\_Tags}_{it-1} + \beta_2 \text{Monthly\_New\_Commits}_{it-1} + \beta_3 \text{OSS\_Experience}_{it-1} \\ & + \beta_4 \text{Developers\_Reputation}_{it-1} + \beta_5 \text{Avg\_Commits}_{it-1} + \beta_6 \text{Project\_Age}_{it-1} \\ & + \beta_7 \text{Project\_Size}_{it-1} + \beta_8 \text{License}_i + \beta_9 \text{Sustained\_Participation}_{it-1} + \eta_i + \delta_{it} \end{aligned} \quad (2)$$

Avg\_Commits<sub>it-1</sub>, as shown in Tables 3 and 4. The reasons for the choices of these moderators are discussed in the control variables section.

## 4 | FINDINGS

Table 3 shows the regression results for initial participation. It was found that both WOM (IP\_Microblogging\_Tags<sub>it-1</sub>)

and OL (Monthly\_New\_Commits<sub>it-1</sub>) positively affect initial participation in both the baseline and interaction models. Project age negatively affects both types of participation. However, it positively moderates the effects of WOM and OL on initial participation. This indicates that talking about or displaying developers' activities of older projects can better attract new developers than younger projects. Moreover, average workload is found to negatively moderate the impacts of OL on initial participation. This



TABLE 3 Regression results for initial participation

| Dependent variable: Initial_Participation <sub>it</sub> |                         |                          |
|---------------------------------------------------------|-------------------------|--------------------------|
| Variables                                               | Baseline                | With interaction effects |
| <b>Independent variables</b>                            |                         |                          |
| IP_Microblogging_Tags <sub>it-1</sub> (WOM)             | 0.73E-03 (1.68E-04)**   | 0.64E-03 (1.99E-04)*     |
| Monthly_New_Commits <sub>it-1</sub> (OL)                | 1.39E-04 (2.11E-05)***  | 1.13E-04 (3.35E-05)***   |
| <b>Control variables</b>                                |                         |                          |
| OSS_Experience <sub>it-1</sub>                          | 3.89E-04 (2.52E-05)***  | 2.96E-04 (1.37E-05)***   |
| Developers_Reputation <sub>it-1</sub>                   | 4.61E-04 (5.12E-05)***  | 3.24E-04 (2.88E-05)***   |
| Avg_Commits <sub>it-1</sub>                             | 2.43E-04 (5.86E-05)***  | 2.02E-04 (4.75E-05)***   |
| Project_Age <sub>it-1</sub>                             | -2.55E-03 (3.57E-04)*** | -1.89E-03 (3.16E-03)***  |
| Project_Size <sub>it-1</sub>                            | 2.37E-04 (2.74E-05)**   | 1.65E-04 (2.44E-05)**    |
| Initial_Participation <sub>it-1</sub>                   | 4.77E-03 (3.69E-04)***  | 4.53E-03 (4.53E-04)***   |
| License <sub>i</sub>                                    | -1.87E-04 (1.21E-05)**  | -1.53E-04 (1.64E-05)**   |
| <b>Interaction terms</b>                                |                         |                          |
| WOM * Project_Age <sub>it-1</sub>                       | —                       | 1.35E-04 (3.02E-05)***   |
| WOM * Avg_Commits <sub>it-1</sub>                       | —                       | 3.87E-05 (4.77E-05)      |
| OL * Project_Age <sub>it-1</sub>                        | —                       | 1.43E-05 (2.72E-06)***   |
| OL * Avg_Commits <sub>it-1</sub>                        | —                       | -4.42E-06 (4.01E-07)***  |
| <b>Number of observations</b>                           | 22,308                  | 22,308                   |
| <b>R<sup>2</sup></b>                                    | 0.44                    | 0.47                     |

Note: The table lists the parameter estimates with standard errors in parentheses.

\* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

means projects that appear to have a higher average workload can discourage developers' initial participation intentions toward these projects.

Table 4 shows that WOM does not significantly affect sustained participation, while OL does. Action (OL) speaks louder than words (WOM) in terms of influencing sustained participation. This result is consistent for both the baseline and the interaction models. Consequently, both project age and average workload do not moderate the impacts of WOM. Similar to initial participation, project age negatively affects sustained participation but positively moderates the impacts of OL. It implies that older projects can benefit more from displaying their activity information in terms of attracting more sustained participation. Interestingly, unlike initial participation, average workload positively moderates the impacts of OL on sustained participation. This means that projects that display more average workload information have more sustained participation. The results of the license variable indicate that developers' sustained participation in a project is negatively associated with its restrictive license choice.

## 4.1 | Differential impacts of WOM and OL on initial and sustained participation

### 4.1.1 | Actions speak louder than words on sustained participation: The accessibility-diagnostics perspective

The main finding is that WOM does not have significant effects on sustained participation. In other words, actions (OL) speak louder than words (WOM) in terms of attracting existing project members to sustain their contribution. On the other hand, both WOM and OL can significantly affect developers' initial participation decisions.

We adopt the information accessibility-diagnostics (IAD) framework (Herr et al., 1991) from marketing literature to offer possible explanations for the above findings. The IAD framework suggests that whether a piece of information will be used to support consumers' adoption decisions largely depends on (a) its accessibility, (b) its diagnostics (perceived relevance), and (c) the accessibility of alternative information. Diagnostics can be defined as the extent to which a piece of information distinguishes among alternative explanations or

TABLE 4 Regression results for sustained participation

| Dependent variable: Sustained_Participation <sub>it</sub> |                         |                          |
|-----------------------------------------------------------|-------------------------|--------------------------|
| Variables                                                 | Baseline                | With interaction effects |
| <b>Independent variables</b>                              |                         |                          |
| SP_Microblogging_Tags <sub>it-1</sub><br>(WOM)            | 1.11E-03 (1.58E-03)     | 1.17E-03 (1.48E-03)      |
| Monthly_New_Commits <sub>it-1</sub> (OL)                  | 3.97E-04 (3.44E-05)***  | 3.90E-04 (3.02E-05)**    |
| <b>Control variables</b>                                  |                         |                          |
| OSS_Experience <sub>it-1</sub>                            | 1.46E-04 (2.27E-05)***  | 1.41E-04 (1.85E-05)***   |
| Developers_Reputation <sub>it-1</sub>                     | 3.38E-04 (4.01E-05)***  | 3.42E-04 (3.07E-05)***   |
| Avg_Commits <sub>it-1</sub>                               | 1.62E-04 (3.51E-05)***  | 0.84E-04 (1.32E-05)***   |
| Project_Age <sub>it-1</sub>                               | -3.53E-03 (2.54E-04)*** | -2.47E-03 (2.05E-04)***  |
| Project_Size <sub>it-1</sub>                              | 2.27E-04 (4.03E-05)***  | 1.75E-04 (2.96E-05)***   |
| Sustained_Participation <sub>it-1</sub>                   | 2.75E-03 (2.95E-05)***  | 1.87E-03 (2.41E-05)***   |
| License <sub>i</sub>                                      | -1.58E-04(1.30E-05)**   | -1.27E-04(1.09E-05)**    |
| <b>Interaction terms</b>                                  |                         |                          |
| WOM * Project_Age <sub>it-1</sub>                         | —                       | 1.07E-03 (2.75E-03)      |
| WOM * Avg_Commits <sub>it-1</sub>                         | —                       | 2.92E-05 (3.85E-05)      |
| OL * Project_Age <sub>it-1</sub>                          | —                       | 2.34E-05 (3.58E-06)***   |
| OL * Avg_Commits <sub>it-1</sub>                          | —                       | 1.63E-06 (3.04E-07)**    |
| <b>Number of observations</b>                             | 22,308                  | 22,308                   |
| <b>R<sup>2</sup></b>                                      | 0.42                    | 0.45                     |

Note: The table lists the parameter estimates with standard errors in parentheses.

\* $p < .01$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

categorizations. Feldman and Lynch (1988) suggested that a piece of information is diagnostic if it helps consumers assign a product to one cognitive category.

Based on IAD, we suggest that both WOM and OL are effective means that can increase the accessibility of information about an OSS project to its potential participants, thereby increasing the likelihood of such information being used in developers' initial participation decisions when there is little alternative project information.

However, after a developer has participated in a project, more diagnostic information (compared with WOM) becomes available to her, such as her knowledge and understanding of the project, her interactions and collaboration experiences with other project members. Such personal knowledge and experiences are often perceived by this developer to have higher diagnosticity than experiences or opinions expressed by others through WOM mechanisms. Then according to IAD, accessible (WOM) information becomes less useful when more diagnostic information becomes available. This is consistent with our empirical findings that WOM information becomes insignificant for existing project members' sustained

participation decisions. On the other hand, OL information in Open Hub remains diagnostic for existing project members. It could be largely due to that it is often considered to be more objective than WOM information.

#### 4.1.2 | Aging effects

Our finding that project age negatively affects both initial participation and sustained participation is consistent with previous OSS research. Hahn et al. (2008) conjectured that it may be because younger projects may have lower technical barriers for new developers to participate since the code and project structure in these projects are not too complex to comprehend. As projects get older, they are very likely to become more complex and will cost participants much more time and efforts to fully understand the code than younger and simpler projects (von Krogh et al., 2003). Therefore, older projects naturally have less attraction for both initial participation and sustained participation from developers.

Moreover, it was found that project age positively moderates the impacts of OL on both initial participation

and sustained participation. As a project ages, its members may gradually develop shared values and group norms and strive for standards of excellence defined within the project. MacIntyre (1981) defines such a process as social practice, in which virtues are transmitted among developers through socialization and collective work. Developers may choose to sustain their contributions, which can match their senses of the common good (von Krogh et al., 2012). We suggest that over time, such social practice can enhance the impacts of OL information (i.e., other members' contribution activities) on individual project members' sustained contributions.

#### 4.1.3 | The moderating effects of core developers (with high average workload)

The most interesting finding is that while the average workload (i.e., average number of commits per developer) negatively moderates the impacts of OL on initial participation, it has positive moderating effects on sustained participation. The former indicates that potential participants are discouraged by OL information of projects that have a higher average workload. Based on our empirical observations, such projects are often dominated by a relatively small number of core developers who account for the majority of project code contributions. Another scenario is that a project only has a small number of participants who on average are contributing more code than projects with similar code contribution sizes but having more members. In both scenarios, projects with a high average workload are often dominated by a small group of core members (Crowston et al., 2006). We conjecture that OSS developers in general prefer not to participate in a project that was already heavily influenced (if not dominated) by a group of core developers.

We suggest that when OSS developers view the profiles that contain OL information of such projects, they are less likely to be influenced to join. Because they may want to become more impactful and visible by joining projects with no or less established core members. In this way, they are more likely to become core members themselves and enjoy benefits such as visibility and higher community status.

On the other hand, average workload positively moderates the impacts of OL on sustained participation. It may be because that the core project members already form a set of social norms and guidelines for their interactions and code development, as well as expectations on workload and performance over time (Crowston et al., 2006; von Hippel & von Krogh, 2003; von Krogh et al., 2003). These social norms, guidelines, and

expectations can motivate the core members to keep contributing like their peers did as the OL information shows.

## 5 | DISCUSSIONS

Our study has important implications for research on virtual collaboration, social influence, and OSS development. First, although there are a few studies (Dabbish et al., 2012; Tsay et al., 2014a; Tsay et al., 2014b) about the impacts of social media and the visibility of developers' activities on OSS project participation, we are among the first to use the social influence research framework to explore how WOM and OL mechanisms may affect individuals' initial and sustained participation in the setting of open source software development. By choosing to participate in a project team, an OSS developer is essentially choosing to collaborate with a group of geographically distributed individuals who heavily rely on information and communication technologies.

Previous research mainly focuses on the impacts of social influence on individuals' (consumers') adoption (initial participation) behaviors (Hars & Ou, 2002; Lakhani & Wolf, 2005) while assuming that the impacts on sustained participation behaviors are similar. However, individuals' initial and sustained participation behaviors differ in many ways. Therefore, we hypothesize that social influence may have differential impacts on these behaviors. Our empirical analysis has demonstrated the importance of taking into account the differences between initial participation and sustained participation.

Second, our empirical results provide both theoretical knowledge and practical insights on how both WOM and OL work in the context of OSS development and how individual developers interact with and influence each other in terms of their project participation behaviors. Adopting a social influence research framework to empirically study OSS developers' participation behaviors allows us to investigate how project and developer characteristics (e.g., age, workload) may moderate the impacts of social influence. Moreover, our empirical model can be extended by adding other social influence factors when relevant data becomes available. The results may help stakeholders design collaboration platforms that can highlight important social influence information and attract more initial and sustained project participation.

Third, we adopted the fixed effects model for our empirical study. This approach to a large extent can control endogeneity issues in the analysis of observational data extracted from a real-world OSS development platform. The Hausman specification test indicates that our choice of the fixed effects regression model can help us

control the biases introduced by the unobserved project heterogeneity and produce consistent estimation results. In general, this approach can be very promising for IS researchers in developing theories or conducting empirical analyses on complex and dynamic social systems, in which individual heterogeneity cannot be overlooked and longitudinal panel data sets are often used.

## 5.1 | Practical implications

Moreover, our study has important practical implications. Our empirical results can help design more effective social functions, interfaces, and information display strategies for platforms that support OSS project team collaborations, in order to attract more initial and sustained participation. Many major OSS development platforms have implemented social (networking) functions like microblogging. The goal is to leverage developers' social networks to influence or shape their behaviors, especially participation behaviors. In behavior economics, Thaler et al. (2008) developed a novel concept called “nudging,” which utilizes information technologies to design choice environments around an individual for altering her behaviors in a predictable way without changing her economic incentives. They also suggested that one of the most effective ways to nudge is via social influence (2008).

Our empirical findings also have practical implications in attracting developers' initial and sustained participation by leveraging WOM and OL effects differently. First, action (OL) speaks louder than words (WOM), which suggests that WOM and OL information has differential impacts on potential and existing project members. When a developer is viewing the profile web page of an OSS project, the Open Hub platform can tell if she is an existing project member using her development history. If so, our empirical results suggest that it is more effective to use and highlight OL information like monthly new commits than WOM messages.

Second, we also found that OL information about projects with a higher average workload negatively affects initial participation decisions and positively influences existing project members' sustained participation. Therefore, the Open Hub platform needs to remove OL information when potential participants are viewing project profile pages and highlight such information for existing members. This can be achieved by using developers' account and development history information in Open Hub. Third, it was also found that the impacts of both WOM and OL on initial participation decrease as the underlying project gets older. Therefore, when recommending

microblogging messages to potential participants, the platform can set the default display order based on the age of the projects or messages.

These strategies aim to help design an OSS development platform that can tailor the social influence information environments for different developers to shape their initial and sustained project participation.

## 6 | CONCLUSION

This study focused on analyzing the differential impacts of WOM and OL on OSS developers' initial and sustained project participation behaviors. Using panel data from more than 1,700 OSS projects and 7,800 developers, we conducted an empirical study using the fixed effects model. Our results indicated that (a) WOM affects developers' initial participation but does not affect developers' sustained participation, while OL positively affects both; (b) project age positively moderates the impacts of OL on both initial and sustained participation; and (c) average workload negatively moderates the impacts of OL on initial participation decisions but positively moderates the effects of OL on existing project members' sustained participation.

This study contributes to both social influence and OSS research. First, our empirical results have demonstrated the importance of social influence in affecting OSS developers' participation behaviors and analyzed its differential impacts on initial and sustained participation. Second, our empirical models can be applied to study how social influence may affect OSS developers' other behaviors using real-world longitudinal data. Third, our findings and discussions offer practical suggestions to design “nudging” strategies or functions in software development platforms. These functions leverage social influence mechanisms to attract more initial and sustained participation. Last, we adopted a data-driven, econometric analysis approach to study the differential impacts of WOM and OL on OSS developers' initial and sustained participation. This approach can largely control endogeneity issues and has great potential to help researchers develop relevant social influence and OSS theories.

In our future study, we will try to extend our models by incorporating social network theories and mechanisms like social capital and structural holes. One limitation of our analysis is that developers' social network features are not included. Our future study plans to analyze developers' kudo sending relationships to extract their social network features. The empirical models used in our study can be easily adapted to incorporate these features to study their impacts on initial and sustained



participation, as well as potential moderating impacts on social influence.

Moreover, the OpenHub platform lacks an important possible channel of social influence in OSS development communities—mailing list. This data source has been used in many empirical OSS studies. The mailing list is similar to microblogging, through which developers can subscribe to development information regarding the OSS projects they are interested. Such information includes developers' technical discussions, proposed code changes, and bug reports (Mockus et al., 2002), and thus may also socially influence developers' participation. Therefore, we plan to analyze such data when it becomes available in our future study.

Another limitation is that the microblogging service embedded in OpenHub narrows down the audience to those who are likely to have OSS programming experience before. On the other hand, public WOM channels like Twitter and Facebook are more open and accessible to the general public, who may or may not have programming skills. Analyzing data from these public channels can help us better understand how social influence may be affected by the learning desires of individuals who have low or no programming skills. Since many OSS development platforms have allowed developers to display their public social media accounts, our future study plans to collect and analyze data from these channels via web crawling.

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**How to cite this article:** Yang X, Li X, Hu D, Wang HJ. Differential impacts of social influence on initial and sustained participation in open source software projects. *J Assoc Inf Sci Technol*. 2021;72:1133–1147. <https://doi.org/10.1002/asi.24481>